

Sandwich Panel Selection Guide

A decision guide for choosing panel type, core material, thickness, joint logic, insulation level, and procurement inputs before ordering.

GUIDE | Technical Guides | 14 pages | March 2026

Introduction: Why Panel Selection Matters

Panel selection is a structural, thermal, and commercial decision made once and paid for over the building's whole life. The wrong core for the climate, or the wrong thickness for the target U-value, does not announce itself on day one — it shows up as condensation on the underside, as a cold store that cannot hold temperature, or as an energy bill that never matches the design.

Selection errors also propagate into the structure. A panel chosen heavier than assumed overloads purlins; a panel spanning further than its profile allows deflects and unzips its joints. By the time these appear, the building is built.

This guide turns selection into nine ordered steps — from building use to procurement — so that type, core, thickness, joint, skin, and load are all resolved before a purchase order is issued.

Step 1: Define Building Use Category

Start from how the building is used. Use category sets the insulation and fire priorities, which in turn drive core and thickness.

Use Type	Typical Span	Insulation	Fire	Recommended Core
Cold Storage	6-12 m	Very High	Medium	PIR 120-200 mm
Industrial Factory	9-18 m	Medium	Low-Medium	PIR 80-100 mm
Food Processing	6-12 m	High	High	PIR 100 mm + FR facing
Logistics / Warehouse	12-24 m	Low-Medium	Low	EPS 80-100 mm
Sports Hall	12-30 m	Medium	Medium	PIR 80-100 mm
Office / Admin	6-9 m	High	High	Mineral Wool 100 mm

Step 2: Select Core Material

Three cores cover most industrial work. Choose by balancing thermal performance, fire rating, weight, and cost.

Property	PIR	PUR	Mineral Wool
Thermal per mm	Best	Very good	Moderate
Fire performance	Good	Lower than PIR	Best (non-combustible)
Weight	Light	Light	Heavy
Relative cost	Higher	Medium	Higher (thickness)
Best for	Cold / efficient envelopes	Cost-sensitive insulation	Fire-critical zones

- PIR (Polyisocyanurate): best thermal per mm, good fire, higher cost.
- PUR (Polyurethane): similar to PIR, slightly lower fire performance.
- Mineral Wool (Rockwool): best fire rating, heavier, bulkier.

Step 3: Determine Required Thickness

Thickness follows the target U-value for the climate zone. Lower U-value (better insulation) needs more thickness; mineral wool needs more thickness than PIR for the same U-value.

Climate Zone	Target U-Value (W/m²K)	PIR Thickness	Mineral Wool
Hot-Dry (Iran plateau)	0.40-0.50	60-80 mm	100-120 mm
Hot-Humid (Khuzestan)	0.35-0.45	80-100 mm	120-150 mm
Cold (Northwest Iran)	0.25-0.35	100-140 mm	150-200 mm
Temperate (Isfahan)	0.35-0.45	80-100 mm	120-150 mm

Values are indicative for early selection; confirm against the project energy calculation.

Step 4: Roof vs Wall Panel Differences

Roof and wall panels are not interchangeable. They differ in slope, joint, drainage, load direction, and fixing.

Aspect	Roof Panel	Wall Panel
Slope	Minimum slope required ($\geq 3^\circ$)	Vertical / horizontal, no slope
Joint type	Weather-lapped, sealed	Tongue & groove / concealed
Drainage	Sheds water along profile	Not a drainage surface
Load direction	Gravity + uplift	Wind in / out
Fixing	Through-fix or concealed clip	Concealed where possible
Facing options	Profiled / trapezoidal top	Flat, micro-rib, or lined

Step 5: Joint Logic Selection

The joint controls weather-tightness and erection speed. Match it to exposure and finish.

Joint	When to Use	Advantage	Limitation
Tongue & groove	General wall / roof	Simple, fast, economical	Visible fastener on some types
Ship-lap	Exposed, weather-critical	Better water shedding	Slightly slower to set
Concealed clip	Architectural facades	No visible fixing	Higher cost, careful setting-out

Cost rises from tongue & groove to concealed clip; choose the simplest joint that meets the exposure and appearance requirement.

Step 6: Skin Material & Coating Selection

The skin and its coating set corrosion life. Match the skin to the environment, not just the budget.

Skin	Corrosion Resistance	Cost	Typical Use / Environment
Galvanized steel	Moderate	Low	Dry interior, sheltered
Galvalume	Good	Medium	General industrial, exterior
Aluminium	Very good	Higher	Coastal, humid, corrosive
Stainless	Excellent	Highest	Chemical, food, marine

Step 7: Structural Load Check Inputs

Before final selection, gather the inputs the panel must be checked against. A panel is only correct relative to its span and loads.

- Dead load — panel self-weight (varies with core and thickness).
- Live load — maintenance access and, where relevant, snow.
- Wind uplift — location-specific, from the wind map / code.
- Span between purlins — drives profile and thickness.
- Deflection limit — typically $L/200$.
- Point loads — services suspended from the panel, if any.

Step 8: Panel Connection to Structure

Most envelope failures are connection failures. The load path runs from panel to structure through the bracket and fastener:

Connection sequence

1. Purlin / structural support set to line and level.
2. Bracket or cleat fixed to the purlin.
3. Thermal break placed between bracket and panel.
4. Panel set and engaged at the joint.
5. Fastener installed to specified torque and spacing.
6. Flashing closes and weatherproofs the junction.

Common connection errors

- Missing thermal break — cold bridge and condensation.
- Wrong fastener type for the skin — corrosion or pull-out.
- Over- or under-torqued fasteners — crushed core or loose fixing.

Step 9: Procurement Checklist Before Ordering

Item	Pass ✓	Fail ✕	Notes
Panel type confirmed (roof / wall).			
Core material confirmed.			
Thickness confirmed against U-value.			
Facing / skin material confirmed.			
Color confirmed with RAL code, in writing.			
Coating / finish confirmed.			
Joint type confirmed.			
Panel lengths confirmed from the schedule.			
Length tolerance reviewed with supplier.			
Quantity with 3% overage included.			
Flashings listed and quantified.			
Gutters and downpipes listed.			
Fasteners specified and counted.			
Sealant type and quantity included.			
Closures / fillers included.			
Lead time confirmed.			
Delivery access reviewed.			
Storage area prepared.			
Delivery sequence agreed.			
Engineering sign-off obtained.			

Common Mistakes & How to Avoid Them

Mistake	Consequence	Prevention
Undersized thickness	Fails U-value; condensation	Confirm thickness vs climate target
Wrong core for fire zone	Fire non-compliance	Match core to fire priority (Step 1)
Missing thermal break	Cold bridge, condensation	Specify break at every fixing
No overage ordered	Shortage mid-install	Add 3% panels, more on accessories
Wrong fastener for skin	Bimetallic corrosion	Match fastener to skin material
Ignoring panel weight	Purlin overload	Check self-weight in load case

SIPANEL Panel System Overview

SIPANEL designs and supplies the full industrial building envelope:

- Standard sandwich panel systems — roof and wall.
- Standing seam ZIP roof system — long-run, low-slope, concealed fix.
- Aluminium cladding facade system — architectural, hidden-fix.
- Daylighting / transparent roofing — polycarbonate and GRP.

For project-specific selection support — core, thickness, joint, and load — contact SIPANEL engineering: info@sipanelco.ir | +98 313 675 1101

Notes & Sign-off

Project Name	
Building Use	
Selected Panel	
Core / Thickness	
Prepared by	
Date	
Notes	

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